



POLICY BRIEF

Into the future: How will the management of Autonomous Weapon System develop?

Anusha Maharjan



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SCHWARZMAN
SCHOLARS

清华大学苏世民书院

KEY POINTS:

- *There is secrecy, ambiguity and incoherency in defining autonomous weapons systems with various emerging challenges such as ethical concerns, humanitarian risks and its legal complications.*
- *Active collaboration, regular discussions and conversations is required now on redefining the AWS resolution; whether regulation or prohibition or a combination of both is needed to solve the arising uncertainties and mistrust.*
- *Human control and adequate legal compliances in the life cycle of AWS can be instrumental in its management.*

EXECUTIVE SUMMARY

The emerging concept of Lethal Autonomous Weapon Systems (LAWS) or Autonomous Weapon Systems (AWS) is a challenging and difficult domain to govern, regulate and legitimize. Various conventional chemical and nuclear weapons pose an existential threat, but they have been limited by treaties such as the Convention on Certain Conventional Weapons (CCW), International Humanitarian Law (IHL), and the Group of Governmental Experts (GGE) based on deterrence theory. LAWS/AWS vary from any known conventional weapon in that their use results in mutually assured destruction, whilst the principle of deterrence does not apply to LAWS; they can be stealthy, rapid, lethal, and the initial attack can be untraceable. LAWS pertaining to its low manufacturing costs entering into the hands of non-state entities heightens the amount of threat. Despite the increased risk of LAWS, there are no clear forms of convention that appropriately define, prohibit, or regulate these weapons of destruction. Because of the diverse growing nature of Artificial Intelligence (AI) employed in LAWS, there are numerous humanitarian, technological, legal, and ethical challenges to establishing a moratorium. The barrier in legalization or regulation of LAWS is in its fluid nature equally weighing in on its useful application to the human welfare as well as its capacity to create a negative and a destructive outcome. Similarly, autonomous weapons have not fully been developed and deployed en masse hence creates the barriers of futuristic regulations.

As such the issues and challenges of autonomous weapons has recently been and will continue to be a cornerstone in debate on how to manage it to ensure that both the ethical and legal requirements whilst not hampering the scientific research and development. The policy brief outlines the main issues that have arisen in this context and provides a number of potential policy options for the future of autonomous weapons which could be applied on a national as well as global scale.

CONTEXT

The use of technology has always revolutionized conventional warfare, but with the introduction of AI, a new paradigm in warfare has emerged known as the human less warfare, which is directly envisioned to limit boots on the ground, reducing military casualties, but its proliferation also increases unprecedented arms race between different nations. The usually accepted definition of autonomous weapons that we discuss in the policy brief involves the ability to choose (detect, inspect, identify, and scan) and neutralize (attack, destroy, and harm) targets without the need for human intervention (*Views of the ICRC on Autonomous Weapon Systems, 2016*). These weapons have been given various names such as Killer Robots or Slaughter Bots. In the past there have been numerous reports of autonomous weapons being utilized for surveillance, targeting, and direct military involvement in various regions of the world. The Samsung SGR 1, for example, is a semi-autonomous weapon deployed on the border between North and South Korea that can recognize, detect, and fire at intruders (*Jacobson, 2017*). Similarly, the Israeli Harpy drone employs high explosive warheads known as 'fire-and-forget' that can hunt down and destroy targets instantaneously (*AI Weapons Are the Third Revolution in Warfare - The Atlantic, n.d.*).

As of 2020, there have been four instances of the autonomous weapons being deployed in military operations or are contracted for future use. The STM Kargu-2, a Turkish drone made against Libyan National Army forces in 2020 could form swarms, has face recognition technology and remains fully operational even if GPS and radio signals are jammed. It is the first ever recorded autonomous weapon used for real military operations. In 2021, Australian Jager-C is another contracted AWS to be used in military operations. Similarly, the controversial US Air Force MQ-9 Reaper developed in 2020 with enhanced artificial intelligence came into the headlines for its smart decision-making capacity (*Jones, 2021*). These rapid developments of AWS continue to pose serious short and long term implications and future challenges. With the development and deployment of AWS it will be easier and quicker to wage wars that are depersonalized resulting in greater humanitarian crisis and worse case scenario - human extinction (*Klincewicz, 2015*). However, the AWS management stakeholders are now tasked with even more complicated decision making that needs to be addressed soon and they are related to ethical, legal and research and development conundrums.

EXISTING POLICY APPROACHES

Despite multiple GGE meetings in 2017, 2018, 2019, and 2020, there has been no clear demarcation on the concepts of autonomous, semi-autonomous, or fully autonomous weapons with various technical and legal challenges, as even the most basic of weapon systems may be technically sophisticated and legally unaccounted for. Furthermore, there are continuing disagreements in categorizing LAWS in accordance with IHL, with many ethical quandaries as no one understands how autonomy may affect the rules of war such as the principle of distinction, proportionality in attack, and the precautions measures to be utilized (The Convention on Certain Conventional Weapons – UNODA, n.d.).

The most recent Article 36, The GGE summit, which took place from August 3–13, 2021, did formally aid in structuring the discussion around prohibition and regulation debate, despite its limitations in formulating a legislative response and prohibiting technologies that target human lives. As a result, there were two opposing factions. The first was composed of progressive states that advocate positive obligation, prohibition, and legal action against all existing AWS. Second, a group of states that accept the fundamental framework of positive obligation and additional regulations to keep control of this system but refuse to take legal action. After years of futile negotiation, reaching two incremental steps of considering regulation and/or ban is a tiny accomplishment, but it does not completely resolve any issues and instead demonstrates a long, convoluted route ahead. Since there is no middle ground between advocating for an outright ban and pursuing legal action against individuals who support positive obligation and regulation, there is unending discussions on these two fronts and will continue in the future as we envision the management of AWS. It is still impossible to comprehend the full extent of development and deployment of AWS due the ambiguous use of its definition, prolonged discussions and the surrounding secrecy around it. The policy brief thus explores three main challenges followed by policy options: definitions and ethical concerns of AWS, development of AI with humanitarian risks and legal challenges.

FUTURE POLICY PRIORITIES

Unlike other global issues the challenges posed by AWS is a recently emerged and novice topic but with the development of technology the future with AWS in military and other civilian use is eminent hence it is pertinent to make efforts to solve the challenges posed by AWS better now than later. The following policy priorities come to the fore as most salient in the coming years:

- Policy interventions should concentrate on **defining autonomous weapon systems** in concrete terms and **strengthening nation states' positive obligations** on the use of AWS in response to the ambiguity in understanding AWS and its ethical implications. To prevent indiscriminate deaths or human suffering at the hands of unmanaged sensors and software, policies should guarantee the full disclosure of the intent and manner of use of AWS and the state's responsibility to oversee them.
- Because of the security concerns of AWS policy making should comprehensively focus on the emerging discussion of **prohibition with legal response versus regulation with stringent control**. These two are at the center of debate as certain nations, NGOs, and other campaigns call for the outright prohibition and banning of all AWS, while other nations emphasize the importance of AWS in research and development. Without considering whether there might be a middle ground, these two dimensions have been met with diametrically opposed viewpoints.
- Use of AWS crosses many different policy domains, all of which have the potential to influence one another. These domains include: artificial intelligence and technology policies, humanitarian law, legal domains, and international politics. Regarding the use, legality, and moral dilemmas of AWS, there must be **coherence** across all policy sectors. In reality, as we have seen in the GGE meetings, Article 36 and IHL refer to LAWS and its detrimental effects, but different governments actively researching in AWS, as well as government organizations like the Pentagon and the Air Force, as well as other technological companies, interpret these bylaws in their own ways arbitrarily or ambiguously to justify their actions.

- **Monitoring, follow up and transparency can help build trust** in the international community. Since AWS are novel, we do not really fully understand how they will evolve and behave in the future. We also don't know how or when they will become lethal because lethality is not innate and depends on intent. Given the myriad ethical, legal, and technical ambiguities, how can one restrict or prohibit something that has not yet reached its full potential? Due to these ambiguities and uncertainties, there is a great deal of mistrust between states who support prohibition and those that manufacture AWS. The research, development, and testing of AWS are all shrouded in military and other secrets, and there are no organized monitoring or detection procedures to encourage accountability. Hence, a suitable channel for AWS monitoring needs to be built, actual follow-up steps in conformity with the law should be implemented, and the decision to report should fall under the purview of both the state and the individuals.
- **Collective responsibilities of states** to internally reduce the risk of humanitarian crisis as a result of wrongful use of AWS can be considered one of the most fundamental principles in AWS management. Collective effort is needed to check in with the advancement of technology, LAWS could now be built for less than \$1,000 by an experienced enthusiast, with parts accessible for purchase online and all open-source technology available for download. This heightens the danger of unregulated LAWS that are intelligent, precise, and less expensive slipping into the hands of non-state actors who can inflict devastation such as political assassination, the use of Ariel drones to perpetrate genocide, and the annihilation of half a city for less than \$10 million (*AI Weapons Are the Third Revolution in Warfare - The Atlantic, n.d.*). The availability of low-cost LAWS manufactures and possibility of increased terrorism without the need of contingent or loss of battalion lives can result in rampant world scale conflicts, armaments proliferation and humanitarian crisis. Smaller nations now have both the capability and confidence to compete in the arms race, thanks to technological advancements and increased production of LAWS. Thus, it has become ever more crucial to check in on these national loopholes and amend them before a crisis.
- **Human element needs to be at the heart of deployment of AWS** or any new advancement in this area. Humans should be at the back end of any military or war related deployment and there should be a clear line of judgment, context and ability to decide unlike the fully autonomous robots capable of making smart decisions being extensively researched on and deployed to save time and resources. The trend of depersonalization of war via AWS needs to be limited.

The three main fundamental policy considerations that should guide the future of AWS management are: Can we? Ought we? Should we? Simply in other words can we implement this new approach in terms of technical and financial aspect, how realistic is it? How specific and attainable are the demanded requirements in management of AWS? Ought we, according to the legal framework implement and or enforce the newer approaches so as to better facilitation and management of AWS and will nation states comply with the legal or stronger policy responses. Finally, should we ethically and morally implement the above-mentioned policy approaches. Will application of these comply with humanistic value? Will they serve for the betterment of human civilization as a whole or are there any loopholes? Are we infringing the rights of future research and development of AWS and only focusing on the negative implications? Thus, these should be the leading questions we can ask ourselves while we implement broader approaches to AWS management nationally and globally.

POLICY RECOMMENDATIONS

The policy options provide us with broad approaches, but concrete action needs to be taken in order to support the proper implementation and management of AWS.

Recommendation 1: Additional resources need to be allocated to design relevant mechanisms that concretely defines autonomous weapon systems. While defining and conceptualizing the term for larger understanding may be the first step but is not the last one. Different nation states, government agencies and manufacturing companies have a different interpretation of AWS, it is crucially important to align them all and remove all the incoherency, simplicities and anthropomorphism in the languages and mandates used for LAWS/AWS. In addition, there is a high need for collaboration between alternative and regional bodies such as the UN, NATO, EU and AU working together to establish an international standard for AWS addressing its technological, legal, ethical and humanitarian challenges.

Even more realistically, the two UN bodies, namely the Committee on Disarmament and International Security and the General Assembly, need to provide a participative platform for discussions and debates more frequently, given the constantly shifting nature of AWS. The concept of AWS is new and evolving, and there have been far less debates of it than other major global issues like poverty and climate change.

Recommendation 2: Legal accountability and Human Control needs to be reached. The IHL, CWC and GGE do not have concrete regulation towards LAWS, and hence there is constant worry that an incoherent legal framework or an absence of it causes an escalation of unpredictable AWS. Thus, the international community needs to focus on prohibition of these unpredictable AWS that are capable of indiscriminate human killings and legally regulate the non-prohibited AWS. Meaningful human control should be maintained across the entire lifecycle of LAWS, from design to development to deployment by manufacturing companies and or nation states. Stakeholders should not be allowed to determine or limit human control/involvement/judgment. Human control and judgment should be non-negotiables. Concerning the legal dimensions of AWS, states should enact national legislation governing state responsibility and product liability to ensure that the state, humans operating it, programmers, corporations, and manufacturers of AWS are held accountable in the event of an error, as responsibility to comply cannot be ever requested from a machine. Before implementation, comprehensive testing, training, and governance should be established. Similarly, a special domestic unit that tightens restrictions on open purchasing of AWS parts through licensing needs to be set up in a way that it also does not interfere with peaceful civilian applications and R&D of autonomous systems.

Recommendation 3: Open mechanism for transparency of AWS development and identifying perpetrators who commit AWS related violations need to be set up. This will hold nations and individuals accountable and vindicate that non-use practice will not go unpunished. State's obligation to comply with transparency in AWS development, unlike the issue with nuclear weapons, can help reduce secrecy around it and deter the illegal and devastating use of AWS that may result in a humanitarian crisis. Similar to this, identifying the perpetrator is the first step, and actively apprehending them while following due process is a crucial part of taking legal action to employ regulation of unpredictable AWS. Thus, it is imperative that increased monitoring, follow-up, media use, and other accountability measures be implemented without hesitation.

CONCLUSION

The preceding policy recommendations do not guarantee an overall solution to the existing AWS concerns, but rather present a realistic approach on three fronts: attempting to define autonomous weapons and ways to reach active collaboration, push for legal accountability, coherency and center the human element with the issues posed by LAWS/AWS. These proposals are provided as newer insights that are realistic and attainable with adequate policy backing and prudence. If we as policy makers in government/academia and other influencing sectors care for the safety and security of civilian population and peaceful growth of technology, then it is necessary to have the care and consideration regarding the advancement of AWS. There is no one right answer: either to prohibit AWS or to regulate them? If chosen one of the options, how do we advance on the decision? Answering these questions is difficult but we need to address them in some fashion. No one wants to see the rise of “Skynet” or “Terminators” dominating earth shown as reality in many Sci-fi movies but hopeful that the policy brief can be taken seriously by relevant agencies, manufacturers and nation states rather than one too many recommendations gone unnoticed.

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